

Problem 6

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Problem 1 A company is producing cell phones. The cost of producing x cell phones is given by $C(x)$, defined by

$$C(x) = -0.01x^2 + 40x + 400, \text{ for } 0 \leq x < 1000.$$

AVERAGE cost of producing the first x cell phones is given by

$$C_{AVG} = \frac{C(x)}{x}$$

If the company has produced x cell phones, the cost of producing one more item is given by

$$\text{COST of producing one more item} = C(x+1) - C(x)$$

MARGINAL COST is approximation of the cost of producing one more cell phone

$$\text{MARGINAL COST} = C'(x)$$

- (a) Compute the average cost of the first 300 cellphones that the company produces.

Explanation. The first 300 cell phones costs, on average,

$$C_{AVG} = \frac{C(300)}{300} = 38.33$$

- (b) Compute the cost of producing one more cell phone, if the company has produced 300 cell phones.

Explanation.

$$C(301) - C(300) = -0.01(301)^2 + 40(301) + 400 - (-0.01(300)^2 + 40(300) + 400) = 33.99$$

- (c) Compute the marginal cost, if 300 cell phones have been produced.

Explanation.

$$\text{MARGINAL COST} = C'(300) = [0.02x + 40]_{x=300} = 34$$

- (d) *Why is the marginal cost a good approximation of the cost of producing one more item? Explain!*

Explanation. By the definition of the derivative we have that

$$C'(300) = \lim_{h \rightarrow 0} \frac{C(300 + h) - C(300)}{h} \approx \frac{C(300 + h) - C(300)}{h}$$

for any h near 0. In particular, for $h = 1$, we obtain

$$C'(300) \approx \frac{C(300 + 1) - C(300)}{1} = C(301) - C(300),$$

Therefore, marginal cost at $x = 300 \approx$ the cost of producing one more item!